Arrays — Complete DSA Interview Preparation

All major array patterns covered | ~130 problems | From Easy to Hard | LeetCode + GFG

1. ARRAY BASICS (8 problems)

■ Must know before anything else

Core Basics

- ■ Find Max / Min in Array
- ■ Reverse an Array
- ■ Rotate Array Left / Right by K
- ■ Remove Duplicates from Sorted Array
- ■ Move Zeros to End
- ■ Find Second Largest Element
- ■ Missing Number (1 to N)
- ■ Check if Array is Sorted

2. HASHING (10 problems)

■ HashMap / HashSet — O(1) lookup pattern

HashMap / HashSet

- ■ Two Sum
- ■ Two Sum II — Sorted Array
- ■ Contains Duplicate
- ■ Contains Duplicate II (within K distance)
- ■ Group Anagrams
- ■ Longest Consecutive Sequence
- ■ Majority Element (Boyer-Moore)
- ■ Find All Duplicates in an Array
- ■ Subarray Sum Equals K
- ■ Top K Frequent Elements

3. PREFIX SUM (8 problems)

■ Build prefix array first, then answer queries in O(1)

Prefix Sum Pattern

- ■ Range Sum Query — Immutable
- ■ Subarray Sum Equals K
- ■ Product of Array Except Self
- ■ Find Pivot Index
- ■ Contiguous Array (equal 0s and 1s)
- ■ Maximum Size Subarray Sum Equals K
- ■ Number of Subarrays with Bounded Maximum
- ■ Count of Range Sum (Hard) ■ Hard

4. TWO POINTERS (12 problems)

■ Opposite ends or same direction — O(n) instead of O(n^2)

Classic Two Pointer

- ■ Two Sum II — Sorted Array
- ■ 3Sum
- ■ 3Sum Closest
- ■ 4Sum
- ■ Container With Most Water
- ■ Trapping Rain Water (Hard) ■ Hard

Same Direction / Fast-Slow on Array

- ■ Remove Duplicates from Sorted Array
- ■ Remove Element

■ Squares of a Sorted Array
■ Sort Colors — Dutch National Flag
■ Merge Sorted Arrays In-Place
■ Minimum Size Subarray Sum

5. SLIDING WINDOW (12 problems)

■ Fixed window vs Variable window — know both types

Fixed Window

■ Maximum Sum Subarray of Size K
■ Maximum Average Subarray I
■ First Negative Number in Every Window of Size K
■ First Smallest Number in Every Window of Size K

Variable Window

■ Longest Substring Without Repeating Characters
■ Minimum Window Substring (Hard) ■ Hard
■ Longest Repeating Character Replacement
■ Subarrays with K Different Integers
■ Sliding Subarray Beauty
■ Maximum Points You Can Obtain from Cards

Deque Based

■ Sliding Window Maximum (Hard) ■ Hard
■ Sliding Window Minimum

6. BINARY SEARCH (14 problems)

■ Apply on sorted arrays OR on answer space — both types critical

Classic Binary Search

■ Binary Search — Basic
■ Search Insert Position
■ First and Last Position of Element in Sorted Array
■ Count Occurrences of Element in Sorted Array
■ Search in Rotated Sorted Array I
■ Search in Rotated Sorted Array II (duplicates)
■ Find Minimum in Rotated Sorted Array

Binary Search on Answer

■ Koko Eating Bananas
■ Capacity to Ship Packages Within D Days
■ Find Peak Element
■ Sqrt(x) — Integer Square Root
■ Median of Two Sorted Arrays (Hard) ■ Hard
■ Aggressive Cows — GFG
■ Allocate Minimum Number of Pages

7. SORTING (10 problems)

■ Know Bubble, Selection, Insertion, Merge, Quick — and their use cases

Sorting Algorithms to Implement

■ Bubble Sort
■ Selection Sort
■ Insertion Sort
■ Merge Sort (Divide & Conquer)
■ Quick Sort (Partition)
■ Counting Sort / Radix Sort

Sorting-Based Problems

■ Sort Colors — Dutch National Flag
■ Largest Number (Custom Sort)

■ Wiggle Sort II
■ Maximum Gap (Hard) ■ Hard

8. MERGE INTERVALS (7 problems)

■ Sort by start time first — then merge/insert

Interval Problems

■ Merge Intervals
■ Insert Interval
■ Non Overlapping Intervals
■ Meeting Rooms I — Can Attend All?
■ Meeting Rooms II — Min Rooms Needed
■ Minimum Number of Platforms — GFG
■ Employee Free Time (Hard) ■ Hard

9. CYCLIC SORT (7 problems)

■ Array contains numbers 1 to N — place each number at index num-1

Cyclic Sort Pattern

■ Cyclic Sort — Basic
■ Missing Number
■ Find All Missing Numbers
■ Find the Duplicate Number
■ Find All Duplicates in Array
■ Set Mismatch
■ First Missing Positive (Hard) ■ Hard

10. SLOW & FAST POINTER (6 problems)

■ Floyd's cycle detection — mainly linked list but applies to array index jumping

Fast & Slow on Arrays

■ Find the Duplicate Number (no extra space)
■ Circular Array Loop
■ Happy Number
■ Find Start of Cycle
■ Middle of Linked List (concept applies)
■ Linked List Cycle II

11. KADANE'S ALGORITHM (5 problems)

■ Max subarray — extend or restart. Master this before DP.

Kadane's Pattern

■ Maximum Subarray — Classic Kadane
■ Maximum Product Subarray
■ Maximum Sum Circular Subarray
■ Longest Turbulent Subarray
■ Best Time to Buy and Sell Stock

12. GREEDY ON ARRAYS (8 problems)

■ Make locally optimal choice at each step

Greedy Pattern

■ Best Time to Buy and Sell Stock II (multiple transactions)
■ Jump Game I — Can Reach End?
■ Jump Game II — Min Jumps
■ Gas Station
■ Candy Distribution (Hard) ■ Hard
■ Partition Array into Disjoint Intervals

■ Task Scheduler
■ Minimum Cost to Connect Ropes

13. DIVIDE & CONQUER (5 problems)

■ Split problem in half recursively

D&C; Pattern

■ Merge Sort
■ Maximum Subarray — D&C; version
■ Count Inversions in Array
■ Median of Two Sorted Arrays (Hard) ■ Hard
■ Closest Pair of Points

14. DYNAMIC PROGRAMMING ON ARRAYS (15 problems)

■ Most frequent in interviews — do NOT skip

1D DP

■ Climbing Stairs
■ House Robber I
■ House Robber II
■ Maximum Subarray — Kadane as DP
■ Coin Change I — Min Coins
■ Coin Change II — Number of Ways
■ Decode Ways

2D DP / Grid

■ Unique Paths
■ Unique Paths II (with obstacles)
■ Minimum Path Sum
■ Maximal Square
■ Triangle — Min Path Sum

Subsequence / Stock DP

■ Longest Increasing Subsequence
■ Best Time to Buy/Sell with Cooldown
■ Best Time to Buy/Sell with Transaction Fee

15. MATRIX / 2D ARRAY (8 problems)

■ Treat matrix as graph for BFS/DFS problems

Matrix Problems

■ Spiral Matrix
■ Rotate Image 90 degrees
■ Set Matrix Zeroes
■ Search a 2D Matrix
■ Number of Islands — BFS/DFS
■ Surrounded Regions
■ Word Search — Backtracking
■ Maximal Rectangle (Hard) ■ Hard

16. STACK-BASED ARRAY PROBLEMS (7 problems)

■ Monotonic stack — next greater / smaller element

Monotonic Stack

■ Next Greater Element I
■ Next Greater Element II (circular)
■ Daily Temperatures
■ Largest Rectangle in Histogram (Hard) ■ Hard
■ Trapping Rain Water — Stack approach (Hard) ■ Hard

■ Sum of Subarray Minimums
■ Remove K Digits

17. BIT MANIPULATION ON ARRAYS (5 problems)

■ XOR tricks — often O(n) with O(1) space

Bit Tricks

■ Single Number — XOR
■ Single Number II
■ Missing Number — XOR approach
■ Find Two Non-Repeating Numbers
■ Power Set using Bitmask

COMPLETE SUMMARY

#	Topic	Problems	Hard
1	Array Basics	8	0
2	Hashing	10	0
3	Prefix Sum	8	1
4	Two Pointers	12	2
5	Sliding Window	12	2
6	Binary Search	14	2
7	Sorting	10	1
8	Merge Intervals	7	1
9	Cyclic Sort	7	1
10	Slow & Fast Pointer	6	0
11	Kadane's Algorithm	5	0
12	Greedy on Arrays	8	1
13	Divide & Conquer	5	1
14	DP on Arrays	15	0
15	Matrix / 2D Array	8	1
16	Stack-Based (Monotonic)	7	2
17	Bit Manipulation	5	0
TOTAL		147	15

Recommended Order: Basics → Hashing → Prefix Sum → Two Pointers → Sliding Window → Binary Search → Sorting → Kadane's → Cyclic Sort → Merge Intervals → Slow-Fast → Greedy → Stack → DP → Matrix → Bit Manipulation

■ Red = Hard problem | ■ = Not done | Target: 3-4 problems/day | Estimated: 7-8 weeks